Credit Risk Modeling for Home Equity Line of Credit (HELOC)

A Predictive Analysis Using GLM, GAM, Quantile Regression, and Neural Networks

1. Introduction & Problem Statement

Home Equity Lines of Credit (HELOCs) are revolving credit products secured by a borrower's home equity. While they offer flexible access to funds, they also expose lenders to the risk of default. Predicting which applicants are likely to become seriously delinquent – defined as 90+ days past due within two years – is essential for responsible underwriting, pricing, and portfolio management.

In this project, we develop and compare several predictive models using the FICO HELOC dataset. Our goal is twofold: (1) to achieve high predictive accuracy, and (2) to derive actionable business insights that can inform credit decisions. The models range from highly interpretable (GLM, GAM) to more complex (neural networks), allowing us to balance transparency and performance.

2. Data Description

- **Dataset:** The FICO HELOC dataset contains 10,459 credit applications, each described by 23 anonymized features derived from credit reports, plus a binary target variable (RiskPerformance). The features include credit utilization, number of trades, delinquency history, recent inquiries, and a consolidated risk estimate.
- **Target Variable:** RiskPerformance takes values Bad (borrower became 90+ days delinquent or worse) and Good (paid as agreed). For modeling, we encoded Bad = 1 and Good = 0. The dataset is slightly imbalanced with 5,459 Bad (52.2%) and 5,000 Good (47.8%).
- **Missing Values:** The dataset uses special codes -7, -8, and -9 to denote missing or condition-not-met values. These were replaced with NaN and subsequently imputed with the column median. The columns with the highest number of missing values were MSinceMostRecentDelq (5,428 missing) and NetFractionInstallBurden (4,007 missing). After median imputation, no missing values remained.
- **Data Split:** The data was split into training (80%) and test (20%) sets using stratified sampling to preserve the original class proportions. The resulting training set contained 8,367 observations (4,367 Bad, 4,000 Good) and the test set 2,092 observations (1,092 Bad, 1,000 Good).

3. Methodology

We employed four complementary modeling approaches, each serving a specific purpose in the analysis.

3.1 Generalized Linear Model (GLM) – Logistic Regression

A logistic regression model was fit using statsmodels to establish a highly interpretable baseline. The model assumes a linear relationship between the log-odds of default and the predictor variables. Coefficients, p-values, and pseudo R-squared were examined to identify significant drivers of risk.

3.2 Generalized Additive Model (GAM)

To capture potential non-linear effects while retaining interpretability, we used a logistic GAM (pygam) with spline terms for a subset of eight features that showed strong significance in the GLM. The GAM produces partial dependence plots that visualize how each feature influences the predicted log-odds, holding other variables constant.

3.3 Quantile Regression

Instead of predicting default probability, we inverted the problem: we modeled the credit score (ExternalRiskEstimate) using the remaining features. Quantile regression was applied at the 10th, 50th, and 90th quantiles to understand how the effect of each feature differs across the risk spectrum. This technique is particularly valuable for risk-based pricing and tail-risk assessment.

3.4 Neural Network

A feed-forward neural network was built using TensorFlow/Keras to capture complex interactions. The architecture consisted of an input layer (23 features), two hidden layers with 64 and 32 neurons (ReLU activation), dropout layers (30%) for regularization, and a sigmoid output for probability prediction. The model was trained with the Adam optimizer, binary cross-entropy loss, and early stopping (patience = 10) on validation loss.

3.5 Model Evaluation

All models were evaluated on the held-out test set using:

- **Area Under the ROC Curve (AUC):** Measures the model's ability to distinguish between Bad and Good applicants.
- **Log-loss:** Assesses the calibration of predicted probabilities; lower values indicate better-calibrated predictions.
- **Calibration curves:** Graphical comparison of predicted probabilities versus actual outcomes.

In addition, the neural network was interpreted using SHAP (SHapley Additive exPlanations) to identify feature importance and direction of influence.

4. Results

4.1 Model Performance

The table below summarizes the performance of the three predictive models on the test set.

Model	AUC	Log-loss
GLM (Logistic)	0.7901	0.5534
GAM	0.7864	0.5594
Neural Network	0.7900	0.5532

All models achieved very similar AUC values around 0.79, indicating good discriminatory power. The GLM and neural network exhibited the lowest log-loss, suggesting slightly better calibration.

4.2 Calibration Curves

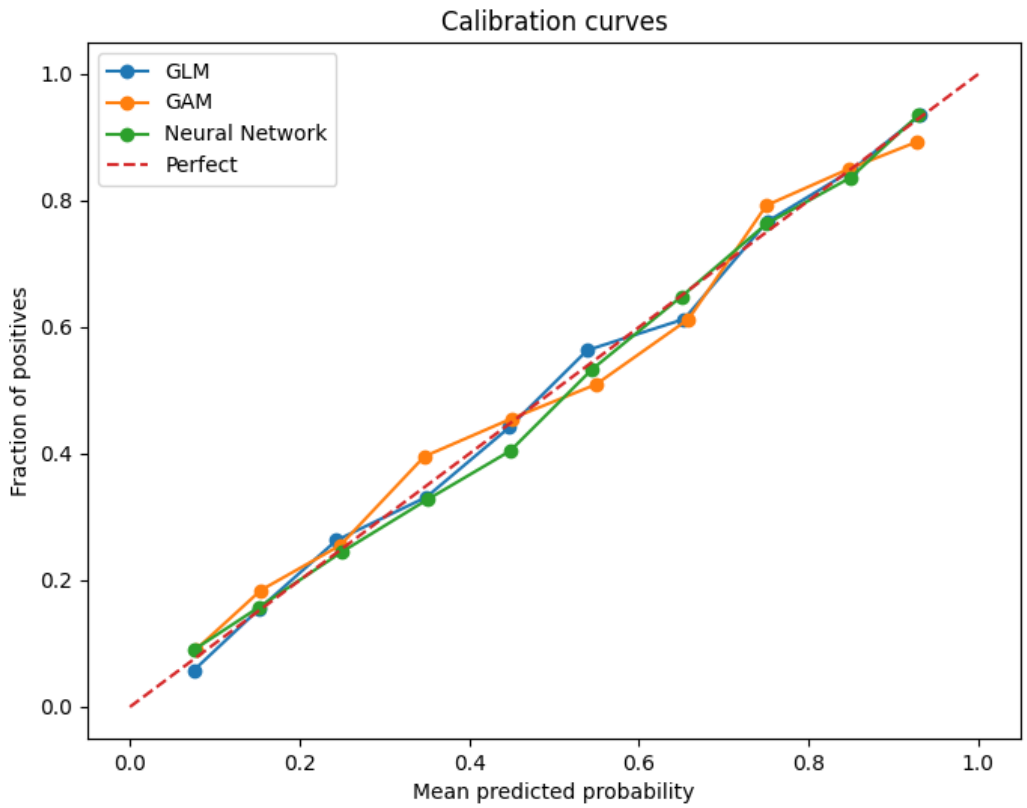

Figure 1: Calibration curves for GLM, GAM, and Neural Network. All three models are reasonably well-calibrated, with predictions close to the diagonal line representing perfect calibration.

4.3 GLM Coefficient Interpretation

Selected coefficients from the logistic regression model are presented below.

Feature	Coefficient	| P>|z|	Interpretation (Odds Ratio)
ExternalRiskEstimate	-0.0532	<0.001	10-point increase → odds multiply by 0.59 (41% lower)
NumInqLast6M	0.3895	<0.001	One more inquiry → odds multiply by 1.48 (48% higher)
NetFractionRevolvingBurden	0.0102	<0.001	10% increase → odds multiply by 1.11 (11% higher)
MSinceMostRecentInqexcl7days	-0.0523	<0.001	More recent inquiry → lower risk
NumRevolvingTradesWBalance	0.0534	<0.001	Each additional revolving trade → odds +5.5%

All coefficients are in the expected direction: higher credit scores and more recent inquiries reduce risk, while higher utilization, many inquiries, and multiple revolving trades increase risk.

4.4 GAM Partial Dependence Plots

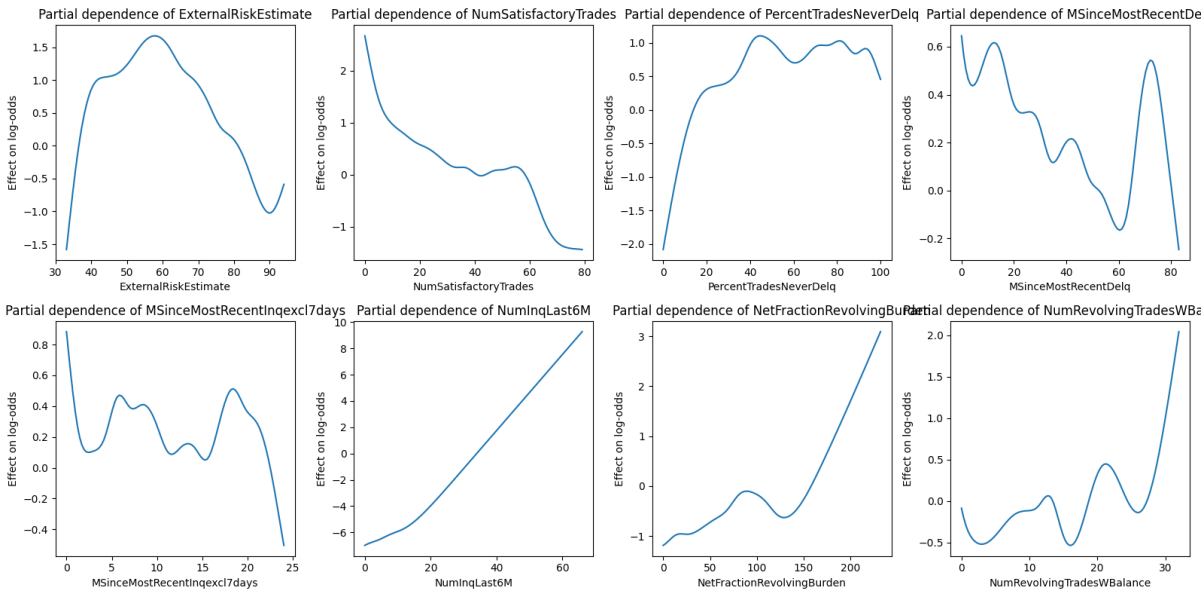

Figure 2: Partial dependence plots for the eight features in the GAM. Non-linear patterns are evident; for example, the effect of NumInqLast6M *flattens after about 4 inquiries, suggesting a threshold beyond which additional inquiries add little incremental risk.*

4.5 Quantile Regression Results

Quantile regression coefficients for selected features at the 10th, 50th, and 90th quantiles of the credit score are shown below.

Feature	q10 (low score)	q50 (median)	q90 (high score)
NumTrades60Ever2DerogPubRec	-1.026	-0.197	-0.331
NetFractionRevolvingBurden	-0.107	-0.113	-0.122
NumRevolvingTradesWBalance	-0.519	-0.349	-0.326
NumInqLast6M	-0.189	0.047	-0.106

Note: All coefficients are negative, meaning higher values of these features lower the credit score, but the magnitude varies by quantile.

The most striking result is for NumTrades60Ever2DerogPubRec: its effect is five times larger at the 10th quantile than at the median. This indicates that past delinquencies are especially damaging for applicants who are already in the high-risk segment.

4.6 Neural Network SHAP Explanation

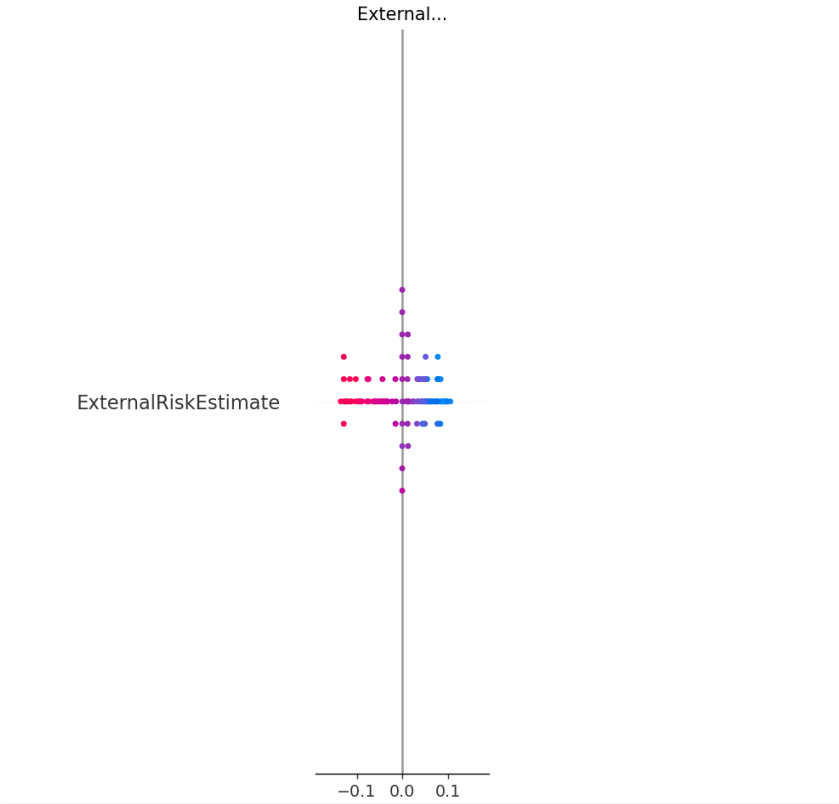

Figure 3: SHAP summary plot for the neural network. Features are ordered by importance (top = most important). Red dots indicate high feature values, blue dots low values. For the most important features, higher values (red) generally push the prediction toward Bad (right side).

The SHAP plot confirms that ExternalRiskEstimate, NumInqLast6M, NetFractionRevolvingBurdenm and MSinceMostRecentInqexcel7days dominate the model's decisions, consistent with the GLM

5. Discussion / Interpretation

5.1 Business Insights from GLM and GAM

- **Credit score is the strongest predictor**: The ExternalRiskEstimate alone can explain a large portion of risk. This aligns with industry practice of using external credit scores as a primary input.
- **Recent credit inquiries signal distress**: Multiple recent inquiries are associated with higher default risk. A rule-based approach could cap the acceptable number of inquiries for automatic approval.
- **Revolving burden matters**: High revolving utilization (credit cards, lines of credit) significantly increases risk. The linear effect in the GLM suggests that even small increases matter, but the GAM shows a slightly nonlinear shape that might be captured by a piecewise-linear function.

5.2 Non-linear Effects (GAM)

The partial dependence plots reveal important thresholds:

- NumInqLast6M: The risk increase is most pronounced between 0 and 2 inquiries; beyond 4, additional inquiries add little extra risk.
- ExternalRiskEstimate: The relationship is nearly linear, but the slope steepens around the lower end, meaning that small improvements in the score for very low-score applicants yield larger risk reductions.

These insights can guide the design of rating factors: e.g., using a spline or categorical binning for inquiries rather than a simple linear term.

5.3 Risk Segmentation via Quantile Regression

Quantile regression provides a powerful lens for risk-based pricing. The finding that delinquent trades have a much stronger impact on low-score applicants suggests that a one-size-fits-all pricing model may be suboptimal. A more nuanced approach could:

- Charge higher premiums for applicants with a history of delinquencies, especially if they already have low scores.
- Offer more competitive rates to applicants who have recovered from past issues but now have high scores.

5.4 Neural Network and Interpretability

Although the neural network offers no direct interpretability, SHAP analysis reveals that its most important features match those from the GLM. This consistency increases confidence in the model's validity. The small gain in AUC (0.7900 vs. 0.7901) may not justify the loss of transparency for regulatory purposes, but the neural network could serve as a secondary "triage" tool for borderline cases where a second opinion is valuable.

5.5 Model Trade-offs

- **GLM**: Most transparent; easy to implement in scorecards; coefficients are directly interpretable. Best for regulatory compliance (fair lending).
- **GAM**: Offers more flexible functional forms while still being interpretable. Good for capturing non-linearities without introducing complexity.
- **Neural Network**: Highest potential for capturing interactions, but requires post-hoc explanation methods (SHAP). Suitable as a supplementary model for automated screening.

6. Conclusion

We successfully developed and compared multiple predictive models for HELOC default risk. All models achieved a similar AUC of approximately 0.79, demonstrating solid predictive power. The GLM provides a transparent baseline with clear coefficient interpretations; the GAM reveals non-linear relationships; quantile regression adds a distributional perspective; and the neural network offers comparable performance with the help of SHAP for explanation.

From a business perspective, we recommend:

- Using the GLM as the primary underwriting scorecard due to its simplicity and regulatory acceptability.
- Incorporating insights from the GAM to refine rating factors (e.g., capping inquiries, using splines).
- Leveraging quantile regression findings to design risk-based pricing strategies that account for differential feature impacts across risk tiers.
- Employing the neural network as a "second look" tool for borderline applications, where a more nuanced assessment is beneficial.

Future work could include collecting more recent data, exploring ensemble methods (e.g., XGBoost), and validating the models on an external dataset to ensure robustness.

7. References

1. FICO (2018). *Explainable Machine Learning Challenge*.
2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
3. Lundberg, S., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.